

9.1 40T - INLINE PIN DRIVE SPINDLE - REPLACEMENT

9.2 40T - INLINE SPIDER COUPLING SPINDLE - REPLACEMENT

9.3 40T - SPINDLE - TROUBLESHOOTING

9.4 40T - SPINDLE DRAWBAR - TROUBLESHOOTING

9.1 40T - INLINE PIN DRIVE SPINDLE - REPLACEMENT

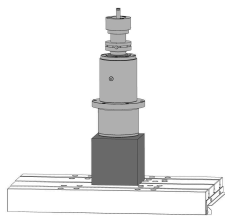
VMC - 40-Taper - Inline Pin-Drive Spindle - Replacement AD0318

Revision B - 12/2025

Introduction

The following service video shows how to replace a 40T Inline Spindle.

 **NOTE:** This service video is for reference only and does not replace the written procedure.



This procedure tells you how to replace a VMC - 40-Taper Inline Pin-Drive Spindle. For a DT/DM machine, refer to the [DT/DM - PIN DRIVE - SPINDLE - REPLACEMENT - AD0650](#) procedure.

This Document Applies to the Following Service Kits:

- **93-30-11972C** SPINDLE, SVC, 40T INLINE 15K PIN DRIVE
- **93-30-11972D** SPINDLE 40T INLINE 15K PIN DRIVE SVC
- **93-30-12298A** SPINDLE 40T INLINE 8/10K PIN W/DB GREASE
- **93-30-12298B** SPINDLE 40T INLINE 8/10K PIN W/DB GREASE
- **93-30-12504A** SPINDLE 40T 8/10/12K VMC INLIN PIN SVC
- **93-30-12504B** SPINDLE 40T INLINE 8/10/12K PIN DRIVE SVC
- **93-30-13432** SPINDLE 40T 8/10/12K VMC INLIN PIN SVC
- **93-30-13432A** SPINDLE 40T 8/10/12K PIN DRV SVC
- **93-30-13455** SPINDLE 40T 15K VMC INCLINE PIN SVC
- **93-30-14228** SPINDLE 40T INLINE 15K PIN DRV W/DB SVC

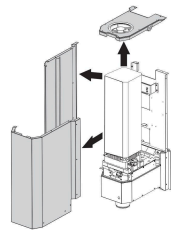
- **93-30-13749** SPINDLE
40T INLINE 8/10K PIN
W/DB GREASE
- **93-30-14456** SPINDLE
40T INLIN 18K PIN DRIVE
W/DB SVC

Note: If you install the same spindle that you remove or a new spindle, record the location of the spindle shims. Install the spindle shims in the same location and then do the [VMC - SPINDLE - SPINDLE SWEEP](#) procedure.

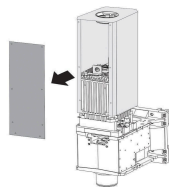
Machine Compatibility

Do this procedure on vertical mills that have a 40-taper inline pin-drive made after May, 2016

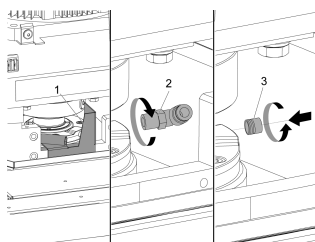
Spindle Removal



1
Remove the spindle head cover panels.



2
Remove the front motor shroud panel.

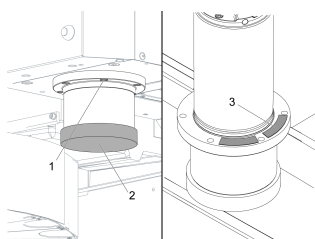


3
Remove the front air scraper [1].

Disconnect the spindle lubrication tube.

Remove the 90° spindle lubrication fitting [2].

Install a 1/8" NPT plug [3] into the lubrication port. Install the plug [3] flush or below flush with the surface of the spindle.



4
Put an oil pan on the table to catch all oil leaks.

Loosen the (6) spindle screws [1] approximately (2) turns.

Install the protective cap [2] over the spindle nose.

⚠ Caution: The protective cap is necessary. It prevents damage to the spindle when it touches the table in the next step.

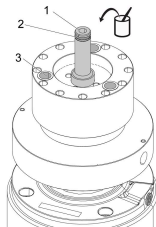
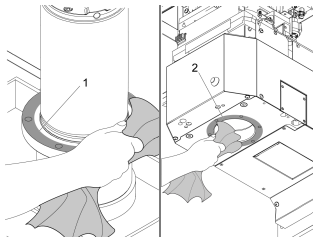
Jog the Z Axis in the negative (-) direction until the spindle touches the table. If the spindle does not touch the table, put a wooden or aluminum block under the spindle.

Remove the (6) spindle screws [1].

Jog the Z Axis in the positive (+) direction until the spindle is fully out of the spindle head and be removed.

Remove the shims [3].

Spindle Installation Preparation



1
Jog the Z Axis in the positive (+) direction to the top of travel.

Clean and stone the spindle flange [1] and the spindle head surface [2] where the spindle flange will touch.

2
Replace the transfer tube [1] if it is worn or damaged.

Apply a thin layer of lithium grease to these parts:

- transfer tube [1]
- transfer tube o-rings [2]
- pin-coupler bushings [3]

Push the transfer tube [1] back into the spindle pin drive coupling.

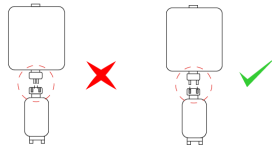
Place the shims in the same location as the previous spindle.

Visually check that three bushings [3] are inside the spindle coupler. Make sure they are the same lengths and aren't damaged.

Spindle Installation

SPINDLE AND MOTOR COUPLING

⚠ Caution: Bearing damage will occur if **ANY** load is placed on the motor shaft or couplings are not aligned during installation.

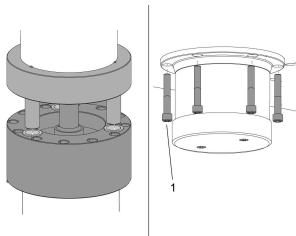


1

⚠ Caution: The motor shaft may become misaligned if spindle installation is done incorrectly.

Before installing the spindle loosen the motor mounting bolts at least a half of an inch.

If the pins are misaligned with the coupler, the motor will visibly lift instead of transmitting all of the axis thrust force into the motor shaft and creates damage on the bearings.



2

Turn the spindle until the oil fitting port [1] points to the front right of the machine.

Slowly jog the Z Axis in the negative (-) direction over the spindle.

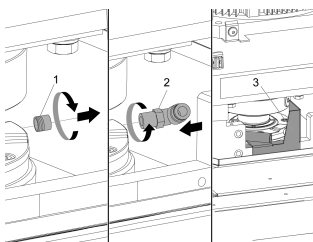
While you jog the Z Axis, use your hand to turn the spindle. If you feel a change in tension, stop immediately. Align the coupler pins to the bushings as closely as possible again.

⚠ Caution: Incorrectly aligned couplers can cause damage to the bearings.

Slowly jog the Z axis in the negative (-) direction.

Make sure the couplers engage smoothly.

Make sure the spindle is centered in the spindle bore. Install (6) spindle screws [1]. Torque the screws. Refer to [HAAS FASTENER TORQUE SPECIFICATIONS](#).



3

Remove the 1/8" NPT plug [1] from the lubrication port.

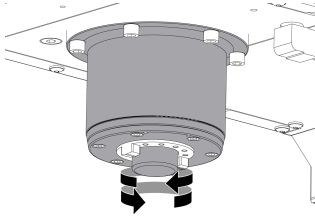
Install the 90° spindle lubrication fitting [2].

Install the front air scraper [3].

Connect the spindle lubrication tube.

Perform the [VMC - SPINDLE - SPINDLE SWEEP](#) procedure.

NOTE: Do not add oil to the spindle casting, even if it had oil before.



4

To align motor to spindle:

Run spindle at 1000 RPM clockwise/forward for 30 seconds.

Run spindle at 1000 RPM counterclockwise/reverse for 30 seconds.

Run spindle at 1000 RPM clockwise/forward for 1 minute.

Snug mounting screws while motor is running.

Note: When motors stops it will jolt to the side and misalign if not snugged down.

Note: The Octavis Vibration Analyzer can be used when aligning motor to find where vibration displacement spike is the smallest.

Torque screws per [HAAS TORQUE SPECIFICATIONS](#).

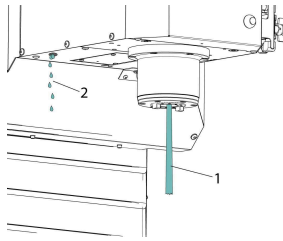
Run the mill spindle vibration test per [MILL - SPINDLE VIBRATION - TOLERANCES](#).

5

- Adjust the Tool Release Piston: Refer to [VMC - 20/30/40-TAPER- TRP - PROXIMITY SENSOR - ADJUSTMENT](#).
- Set Z-Axis Tool Change Offset:
 - Refer to [VMC - SIDE-MOUNT TOOL CHANGER - SET PARAMETER 64 - Z-AXIS TOOL CHANGE OFFSET](#).
 - Refer to [SIDE-MOUNT TOOL CHANGER - Z-AXIS TOOL CHANGE OFFSET - NGC](#).
- Set Spindle Orientation Offset:
 - Refer to [SIDE-MOUNT TOOL CHANGER - SPINDLE ORIENTATION OFFSET - PARAMETER 257](#).
 - Refer to [SIDE-MOUNT TOOL CHANGER - SPINDLE ORIENTATION OFFSET - NGC](#).
- Run the spindle break-in program: [SPINDLE - BREAK-IN PROGRAMS - LATHE AND MILL](#).

Re-install the sheet metal assembly.

Testing

**1**

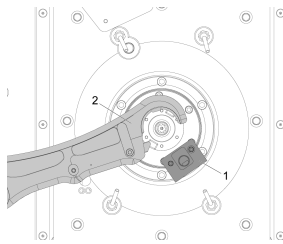
Push **[COOLANT]** to start the TSC.

Correct operation is as follows:

1. Coolant flows correctly from the TSC tool holder [1].
2. A few drops flow from the vacuum-generator hose [2].

Push **[RESET]**.

Make sure there are no leaks in the air hoses and coolant hoses.

**2**

For machines with Accessory Tooling Block (ATB) option:

Install the Accessory Tooling Block (ATB) and verify the tool changer arm clears the block during a tool change.

Note: The image shown is for illustrative purposes only. The ATB position may vary by machine but must not obstruct the tool changer arm during operation.

9.2 40T - INLINE SPIDER COUPLING SPINDLE - REPLACEMENT

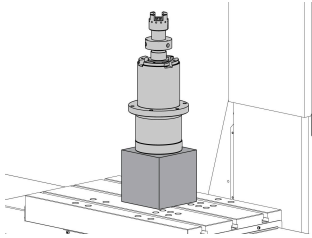
VMC - 40-Taper - Inline - Spider Coupling Drive Spindle - Replacement AD0061

Applies to machines built from: January, 1989

Introduction

The following service video shows how to replace a 40T Inline Spindle.

Note: This service video is for reference only and does not replace the written procedure.



This procedure tells you how to replace a VMC - 40-Taper Inline Spider-Drive Spindle. For a DT/DM machine, refer to the [DT/DM - SPIDER COUPLING DRIVE - SPINDLE - REPLACEMENT -AD0651](#) procedure.

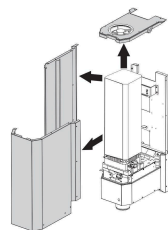
This Document Applies to the Following Service Kits:

- **93-30-12005B** SPINDLE
40T INLINE 12/15K VMC
W/DB
- **93-30-12005C** SPINDLE
40T INLINE 12/15K VMC
W/DB
- **93-30-12008B** SPINDLE
40T INLINE 8/10K VMC
W/DB
- **93-30-4018E** SPINDLE
40T INLINE 12K 2ND GEN
VMC

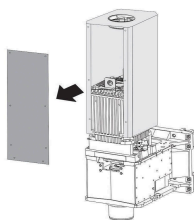
Note: If you install the same spindle that you remove or a new spindle, record the location of the spindle shims. Install the spindle shims in the same location and then do the [VMC - SPINDLE - SPINDLE SWEEP](#) procedure.

Note: If the machine was made between 10/2013 and 2/2016 you must replace the transfer tube.

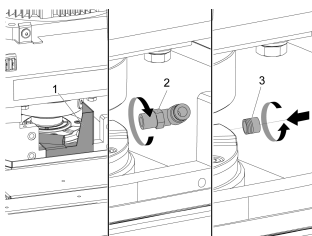
Spindle Removal



1
Remove the spindle head cover panels.



2
Remove the front motor shroud panel.



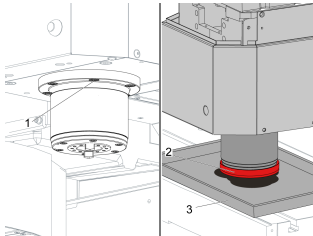
3
Remove the front air scraper [1].

Disconnect the spindle lubrication tube.

Remove the 90° spindle lubrication fitting [2].

Install a 1/8" NPT plug [3] into the oil fitting port on the spindle. Install the plug [3] flush or below flush with the surface of the spindle.

Loosen the (4) bolts that attach the motor to the mounting plate. Do not remove the bolts.



4

Put an oil pan [3] on the table to catch all oil leaks.

Loosen the (6) spindle screws [1] approximately (2) turns.

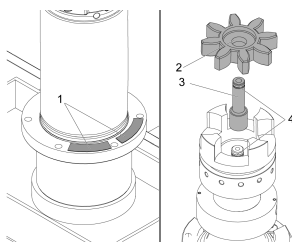
Install the protective cap [2] over the spindle nose.

⚠ Caution: The protective cap is necessary. It prevents damage to the spindle when it touches the table in the next step.

Jog the Z Axis in the negative (-) direction until the spindle touches the table. If the spindle does not touch the table, put a wooden or aluminum block under the spindle.

Remove the (6) spindle screws.

Jog the Z Axis in the positive (+) direction until the spindle is fully out of the spindle head and it can be removed.



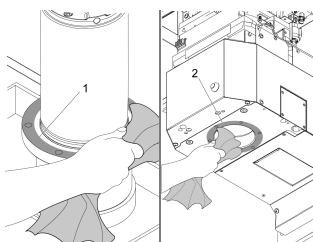
5

Remove the shims [1].

Remove these parts:

- the spider coupling [2]
- the transfer tube [3]
- the o-rings [4]

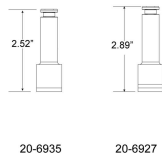
Spindle Installation Preparation



1

Jog the Z Axis in the positive (+) direction to the top of travel.

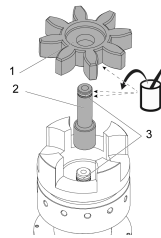
Clean and stone the spindle flange [1] and the spindle head surface [2] where the spindle flange will touch.



2

Use the correct transfer tube for your machine.

- 40-Taper; 8K/10K/12K RPM; inline spindle; use 20-6935: SPINDLE TRANSFER TUBE INLINE 12K
- 40-Taper; 15K RPM; inline spindle: use 20-6927: SPINDLE TRANSFER TUBE INLINE 15K RPM



3

Apply a thin layer of lithium grease to these parts:

- spider coupling [1]
- transfer tube [2]
- o-rings [3]

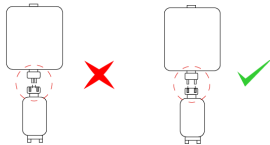
Install the transfer tube into the spindle.

Install the spider coupling.

Place the shims in the same location as the previous spindle.

Spindle Installation

SPINDLE AND MOTOR COUPLING



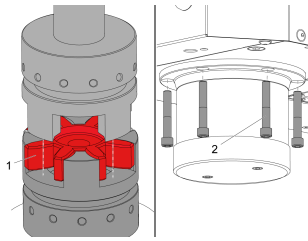
⚠ Caution: Bearing damage will occur if **ANY** load is placed on the motor shaft or couplings are not aligned during installation.

1

⚠ Caution: The motor shaft may become misaligned if spindle installation is done incorrectly.

Before installing the spindle loosen the motor mounting bolts at least a half of an inch.

If the pins are misaligned with the coupler, the motor will visibly lift instead of transmitting all of the axis thrust force into the motor shaft and creates damage on the bearings.



2

Slowly jog the Z Axis in the negative (-) direction over the spindle.

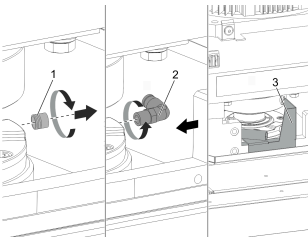
Make sure the spindle coupler and the motor coupler align with the spider coupler [1].

⚠ Caution: Incorrectly aligned couplers can cause damage to the bearings.

Slowly jog the Z axis in the negative (-) direction.

Make sure the couplers engage smoothly.

Make sure the spindle is centered in the spindle bore.
Install (6) spindle screws [2].
Torque the screws. Refer to [HAAS FASTENER TORQUE SPECIFICATIONS](#).



3

Remove the 1/8" NPT plug [1] from the oil fitting port on the spindle.

Install the 90° spindle lubrication fitting [2].

Install the front air scraper [3].

Connect the spindle lubrication tube.

Do the [VMC - SPINDLE - SPINDLE SWEEP](#) procedure.

📄 Note: Do not add oil to the spindle casting, even if it had oil before.

4

To align motor to spindle:

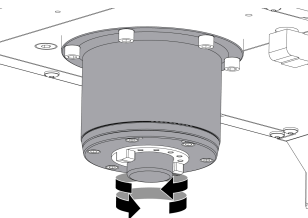
Run spindle at 1000 RPM clockwise/forward for 30 seconds.

Run spindle at 1000 RPM counterclockwise/reverse for 30 seconds.

Run spindle at 1000 RPM clockwise/forward for 1 minute.

Snug mounting screws while motor is running.

📄 Note: When motors stops it will jolt to the side and misalign if not snugged down.



Note: The Octavis Vibration Analyzer can be used when aligning motor to find where vibration displacement spike is the smallest.

Torque screws per
[HAAS TORQUE SPECIFICATIONS](#).

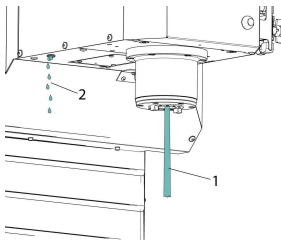
Run the mill spindle vibration test per [MILL - SPINDLE VIBRATION - TOLERANCES](#).

5

- Adjust the Tool Release Piston: Refer to [VMC - 20/30/40-TAPER- TRP - PROXIMITY SENSOR - ADJUSTMENT](#).
- Set Parameter 64 (CHC): Refer to [VMC - SIDE-MOUNT TOOL CHANGER - SET PARAMETER 64 - Z-AXIS TOOL CHANGE OFFSET](#).
- Set Parameter 64 (NGC): Refer to [VMC - SIDE-MOUNT TOOL CHANGER - SET PARAMETER 64 - Z-AXIS TOOL CHANGE OFFSET](#).
- Set Parameter 257 (CHC): Refer to [SIDE-MOUNT TOOL CHANGER - SPINDLE ORIENTATION OFFSET - PARAMETER 257](#).
- Set Parameter 257 (NGC): Refer to [SIDE-MOUNT TOOL CHANGER - SPINDLE ORIENTATION OFFSET - PARAMETER 257](#).
- Balance the spindle: Refer to [SPINDLE - BALANCE INSTRUCTIONS](#).
- Run the spindle break-in program: [SPINDLE - BREAK-IN PROGRAMS - LATHE AND MILL](#)

Re-install the sheet metal assembly.

Testing



1

Push **[COOLANT]** to start the TSC.

Correct operation is as follows:

1. Coolant flows correctly from the TSC tool holder [1].
2. A few drops flow from the vacuum-generator hose [2].

Push **[RESET]**.

Make sure there are no leaks in the air hoses and coolant hoses.

9.3 40T - SPINDLE - TROUBLESHOOTING

Mill Spindle - Troubleshooting Guide

TG0101

Revision B - 11/2025

Introduction

⚠ Download and fill out the Spindle Inspection Report Checklist below before replacing any parts.

SPINDLE INSPECTION REPORT CHECKLIST

Symptom Table

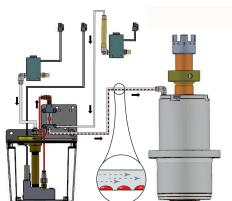
SYMPTOM	POSSIBLE CAUSE	CORRECTIVE ACTION
Spindle not turning and at Power On: A Warning appears "The spindle may be rotating freely without power... "	The software version is outdated.	Check the software version to be 100.20.000.1200 or higher. Contact your Haas Factory Outlet to have the software updated.
	A NGC software error in version 100.17.000.xxx that caused a noticeable growling noise during rigid tapping, particularly with coarse pitch threads.	Update software to the latest version.
Noise during rigid tapping.	Problems with the spindle encoder.	Look for runout, loose mounting screws or shaft clamps, and any other obvious problems. Spindle Non-Contact Encoder (NCE)- Troubleshooting Guide - NGC
	The spindle has been crashed and now damage.	Review alarm history for indications of a crash, check for visual damage to the spindle
The Spindle is seized or it won't turn.	There is heavy side loads causing the cap to gall the shaft.	Check for clearance all the way around with a piece of 0.001" shimstock; loosen the cap, re-center and tighten.
	The spindle lubrication system is not functioning correctly.	Inspect the spindle lubrication system.
	Spindle installed incorrectly	Check Motor coupler for damage or indentations at the pin interfaces. Contact local HFO about issue including relevant pictures of the spindle to motor coupler.
The spindle temperature exceed 140° F (60° C)	The spindle lubrication system does not function correctly.	Check for a clog in the spindle lube pump and verify the type of oil.

The spindle makes an unusual noise.	The spindle drive belt is worn or damaged.	Check the spindle drive belt for damaged.
	The encoder pulleys or drive belt are worn or damaged.	Check the encoder pulleys and belt.
	The spindle lubrication system does not function correctly.	Check for a clog in the spindle lube pump and verify the type of oil.
The surface finish shows chatter, or the spindle vibrates. Alarm 966 EXCESSIVE PART/TOOL IMBALANCE.	The Main Processor PCB standoff hardware is loose or the Control Cabinet mounting hardware is loose or causing a bind.	Tighten the Main Processor PCB standoff hardware. Loosen the Control Cabinet mounting hardware, snug the hardware, then torque down the hardware.
	The tool or tool holder may be too long for the application.	Longer tools are less stable than shorter tools. Always use the shortest tool possible, and remember that a 10% reduction in the length-to-diameter ratio results in a 25% increase in tool stiffness. Unless absolutely necessary, the tool's stick-out from the holder should not be more than 3 times longer than its diameter.
	The spindle drive belt is worn or damaged.	Check the spindle drive belt.
	The encoder pulleys or drive belt are worn or damaged.	Check the encoder pulleys and belt.
	The toolholder is fretting.	Verify the length of the tool with the manufacturer's specifications.
	The drawbar force is incorrect.	Make sure the drawbar clamp force is correct. Refer to: VMC - Spindle - Drawbar - Force Measurement
	The fixture does not support the part.	Adjust the cutting load or program.
	The tool is unbalanced.	Balance the tool holder with the cutting tool in it. If you cannot balance the tool assembly, replace it with one that you can balance.
	Excessive Endmill Runout - Mill.	Tool runout should not exceed 0.0003" (0.076 mm). To check the tool runout, place a dial indicator on the tool, and then rotate the spindle by hand.
	The spindle bearings are damaged.	Perform a vibration analyzer.
	The spindle is not balanced.	Contact your HFO to balance the spindle.

Tools stick or do not come out of the spindle easily.	The tool stayed in the spindle for hours while the machine was not in operation.	Make sure a tool is not in the spindle when the machine is not in operation.
	The volume of air supplied to the machine is not sufficient.	Refer to: New Machine / Pre-Installation information for the correct air requirements.
	The drawbar force is incorrect.	Make sure the drawbar clamp force is correct.
	The toolholder is fretting. Damage tool shank or spindle taper.	Verify the length of the tool with the manufacturer's specifications. Maintain and inspect tool holders, and spindle taper.
	There is no grease on the pull stud.	Lubricate the pull stud.
	(Through-spindle coolant only) Some coolants cause tools to stick.	Coolants containing paraffin waxes can causing sticking tools when using high pressure Through Spindle Coolant with small orifice tooling. Test with a different type of coolant.
	The spindle is getting too hot.	Test the spindle lubrication system. Lack of, or excessive oil will create heat in the spindle.
While running there is bearing Noise: squealing, screeching or skating.	Long gage length tooling.	Stub up the tools to the extent possible. Reduce depth of cut. Use dynamic tool path and constant cutter engagement.
	Noise from imbalance and vibration	Make sure the tools are balanced.
Rubbing noises; Heat at the front cap.	Squealing noises from the TSC union.	Check the TSC union.
	Spindle cartridge has been bumped there is a chance the front cap has been pushed to one side and is rubbing on the nose of the spindle shaft.	Check for clearance all the way around with a piece of 0.001" shimstock; loosen the cap, re-center and tighten.
Noises during tool change.	Some models have had the precharge circuit eliminated.	This is a reliability improvement but a noise can be heard when the TRP contacts the drawbar or the release ring. This is normal.
The spindle oscillates during spindle orientation.	The Wye contactor (if equipped) coil is not receiving power.	Make sure the Wye/Delta contactors are operating correctly. Refer to: Wye-Delta Contactor - Troubleshooting Guide .
	The spindle motor wiring is incorrect.	The spindle motor wiring is incorrect. Refer to: Spindle Motor - Wiring Configurations .
	The vector drive is defective.	The vector drive is defective. Refer to: Vector Drive - Troubleshooting Guide .

Alarm 174 Tool Load Exceeded	Tool Load limit incorrect	View the Advanced Tool Management (ATM) tab to adjust user defined tool load limit
	Aggressive feedrates	Adjust your program for a more conservative feedrate
Machine is experiencing surface finish issues	Preload set to the bearings may be incorrect	Perform the Spindle Deflection Test - Mill procedure for preload verification NOTE: Only perform this procedure <u>AFTER</u> all other troubleshooting steps outlined in the Mill Spindle Inspection Report, linked at the top of this page, have been performed.
Lube line slips out of fitting easily.	Plastic fitting has failed.	Replace fittings with updated steel version. Order PN 58-2196 .

Lubrication



Corrective Action:

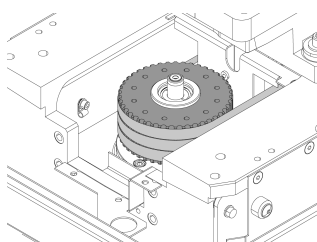
Insufficient or excessive oil supply to the spindle bearings can cause the spindle to overheat. Using the incorrect spindle lubrication oil on the spindle bearings can cause the spindle to overheat. Make sure you use the correct spindle lubrication (SHC 625) and the air pressure to the spindle is correct.

Inspect all tubing, hoses and fittings in the lubrication system for leaks. Check for puddles of oil to help locate leaks.

Go to [HAAS SPINDLE MINIMUM LUBRICATION TROUBLESHOOTING](#) for additional troubleshooting information

Go to [MECHANICAL BIJUR LUBE PUMP - TROUBLESHOOTING GUIDE](#) for additional troubleshooting information.

Drive Belt



Corrective Action:

Power off, lock and tag out machine. Remove the spindle head covers. Disconnect the air supply to the machine. Remove the four 3/8-16 SHCS on the TRP. Remove the TRP and set it aside.

Inspect the spindle drive belts and pulleys for damage:

Look for frayed, stretched or broken belts, and damaged or stripped cogs on the inside surface of the belt. Refer to the [DRIVE BELT - TROUBLESHOOTING GUIDE](#).

Encoder Pulley/Belt

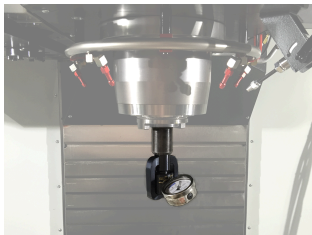
Corrective Action:

Make sure the encoder belt is adjusted correctly and is not damaged. Replace a damaged or worn belt.

Make sure the encoder pulley is not worn. Make sure the set screw is tight.

Reference [SPINDLE NON-CONTACT ENCODER \(NCE\) - TROUBLESHOOTING GUIDE - NGC](#) for more information about the encoder assembly.

Drawbar Force

**Corrective Action:**

Use a drawbar force gauge in the spindle to measure the clamp force. Refer to [SPINDLE - DRAWBAR FORCE REFERENCE TABLE](#).

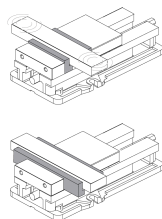
If the drawbar force is incorrect, go to [SPINDLE - DRAWBAR - TROUBLESHOOTING GUIDE](#).

Coolant

Corrective Action:

If the machine has through-spindle coolant (TSC), different types of coolant can cause the toolholder to stick in the spindle. Consider a different type of coolant if the symptom persists.

Workholding



If the workholding does not adequately support the part, it can cause chatter.

Corrective Action:

Stiffen the workholding if possible.

Lower the cutting load with adjustments to the tool selection, or the program.

Balance

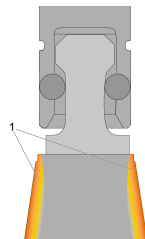
Corrective Action:

Make sure the spindle is balanced. Your local HFO can balance these spindles:

- All belt-driven spindles
- Inline spindles made after August, 2016

All tooling that runs over 10K RPM must be balanced. Do not exceed the manufacturer recommendations for special tooling.

Toolholder Is Sticking



This effect can occur while the machine is in operation if a cold toolholder is placed into a warm spindle taper. This creates thermal expansion [1] on the sides of the toolholder and makes it stick in the spindle taper before it is released.

A common example is if you change from a hot cutter to the cool spindle probe to check your part at the end of a cycle. If the cool probe is left in the hot spindle overnight, it will make a "popping" noise when it is released.

Corrective Action:

Clean the toolholders and spindle taper. Check for damaged toolholders before you put them back in the machine. A damaged toolholder can cause damage to the spindle taper and create tool runout or problems with finish.

Make sure the pull stud is lubricated. Refer to the [MILL - TOOLING - MAINTENANCE](#) procedure.

Toolholder Is Fretting



Corrective Action:

Look for fretting on the toolholders. Long tools can cause fretting on toolholders if you use them aggressively. If there is fretting on the toolholder, or if the symptom

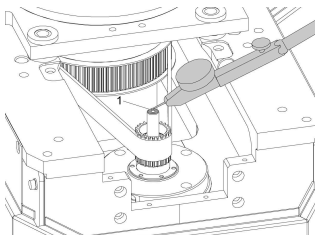
persists, go to [SPINDLE - DRAWBAR - TROUBLESHOOTING GUIDE](#).

Bearings

Corrective Action:

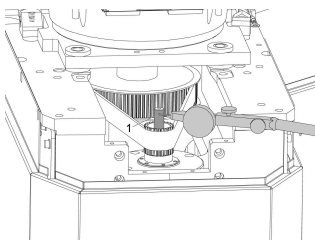
Damaged bearings in the spindle can cause finish issues on the part. Contact your local HFO for a vibration analysis to see if the bearings are damaged.

Axial Check



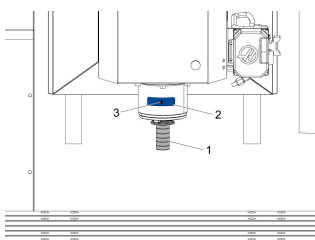
For belted spindles, place the indicator on the top face of the drawbar to check the axial runout.

Radial Check



For belted spindles, place the indicator on the shaft of the drawbar to check the radial runout.

Spindle Temperature Check



1
Note: DO NOT operate the spindle without a dummy tool in the taper [1]. Running the spindle without a tool will damage the bearings.

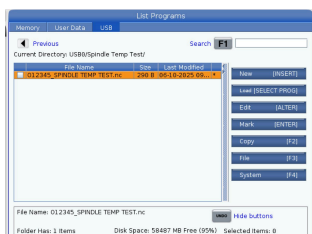
Note: Perform a spindle lubrication test and run spindle warm-up program before proceeding.

Place a piece of masking tape [2] on the front side of the spindle. Draw a black dot [3] in the center of the tape. Take the thermal readings from this location.

2
Download the program listed below onto a USB.

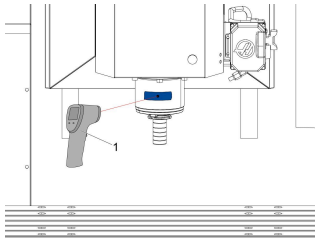
Spindle Temp Test:

012345



Install the USB with the downloaded program into the USB port on the side of the pendant.

Load and run program **O12345**.



3
After the program is done running (40 minute run time), use a temperature gun [1] to measure the spindle temperature.

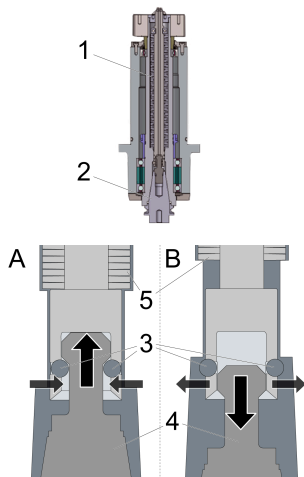
Verify the spindle temperature does not exceed 140° F (60° C).

9.4 40T - SPINDLE DRAWBAR - TROUBLESHOOTING

Spindle - Drawbar - Troubleshooting Guide
TG0132

Revision A - 12/2025

Introduction



- 1. Drawbar
- 2. Toolholder
- 3. Retention balls
- 4. Pull stud
- 5. Belleville springs

Symptom Table

SYMPTOM	POSSIBLE CAUSE	CORRECTIVE ACTION
The toolholder frets in the taper.	The tool pulls out of the taper.	Check for damaged Belleville spring washers.
The part has a poor finish.	The tool length is too long.	Refer to the tool manufacturer's specifications.

The toolholder drops before the tool changer can grab it.	The clamp force on the toolholder is too low, or the TRP pre-charge is not set correctly.	Check the clamp force and pre-charge air pressure.
(TSC Only) Excessive coolant comes out of the coolant tube behind the spindle.	The carbide insert on the drawbar is damaged.	Check the carbide insert in the drawbar and TRP for damage. Note: If the machine has a pre-charge system check that the pre-charge air pressure is set correctly. Refer to the tool release piston pre-charge adjustment procedure: VMC - 30/40-Taper - Tool Release Piston - Pre-charge - Adjustment
The drawbar does not release the toolholder smoothly.	The pull stud angle is pitted or damaged.	Check the pull stud for damage.
	The TRP sensor(s) are not aligned.	Align the tool release piston sensor(s).
	The air pressure or volume to the TRP is too low.	Check the air pressure and volume.
	The drawbar retaining balls are dry.	Apply RED-i grease to the retaining balls.
Toolholder makes a "popping" noise when it is released.	The toolholder and spindle taper are dirty.	Clean the taper.
<u>40T and 50T Belted Spindles</u> Pull studs will not enter into spindle taper, carbide tips on drawbar and TRP are damaged prematurely, And possibly one of the following alarms: Alarm 130 Tool Unclamped Alarm 134 Tool Unclamp Fault Alarm 9970 Unexpected Tool Release	There is a small delay from the I/O PCB that causes the pre-charge solenoid not to turn OFF.	Order service kits 93-3451 & 93-3452 and follow the Low Load Solenoid Valve Kit - Installation procedure.
Service replacement drawbar will not fit into spindle shaft.	Replacement drawbar needs to be straightened before installation.	Roll replacement drawbar along a high-precision granite table to identify bends or waves in drawbar spring. Mark high spots. Clamp drawbar firmly and use a mallet to tap spring back into place. Take care not to damage drawbar. If drawbar cannot be straightened with this method, contact the service department for a replacement.

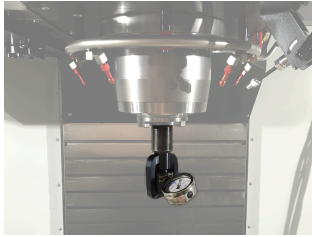
Toolholder Is Fretting



Corrective Action:

Verify that the tool length, axis feed, and spindle RPM are correct according to the tooling manufacturer's recommendations.

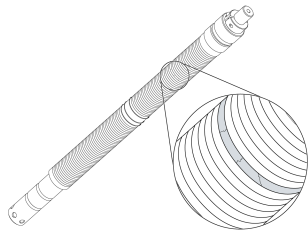
Drawbar Force



Corrective Action:

Check the clamp force with a clamp force measuring tool and refer to the [DRAWBAR FORCE REFERENCE TABLE](#). If the pull force is low, the drawbar is faulty.

Belleville Springs

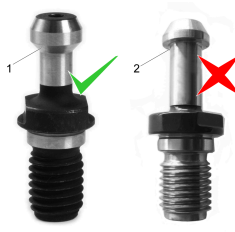


Corrective Action:

If the Belleville springs on your drawbar are damaged, you must replace the drawbar. The drawbar in inline spindles cannot be removed. If the drawbar in an inline spindle is damaged, you must replace the entire spindle assembly.

Check for tools getting stuck in the spindle, inspect the pull studs and spindle taper for damage.

Pull Studs



Corrective Action:

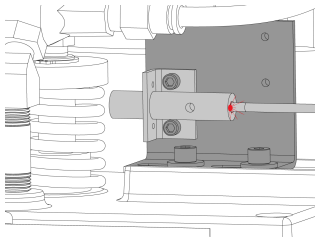
Remove the toolholder from the spindle. Check for damage to the pull stud. The pull stud must have a 45° angle [1].



Ball marks on the edge of the pull stud indicate that the drawbar does not open completely.

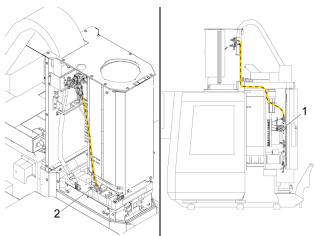
Refer to the [TRP CLAMP/UNCLAMP SENSOR ADJUSTMENT](#) procedure.

TRP Sensor(s)

**Corrective Action:**

Refer to the [TRP CLAMP/UNCLAMP SENSOR ADJUSTMENT](#) procedure.

Air Pressure

**Corrective Action:**

Verify that the air pressure to the TRP solenoid [1] is 90-psi (6.21 bar).

If the volume is correct, there should be only a 10-psi (0.69 bar) air pressure drop during a tool change. Use an inline air gauge to measure the change in air pressure.

Check for kinks or splits in the air hose that goes to the TRP assembly [2].

Make sure the volume of air supplied to the machine is sufficient. Refer to [NEW MACHINE SETUP/PRE-INSTALLATION](#).

Lubrication

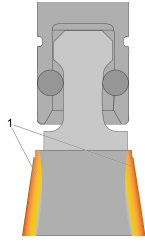
**Corrective Action:**

To grease the retaining balls, put a ball of RED-i grease on top of a pull stud.

Put the toolholder into the spindle. Clamp and unclamp (4) or (5) times to load the grease onto the retaining balls.

Remove the toolholder. Wipe excess grease from the toolholder and spindle taper.

Contamination

**Corrective Action:**

Clean the toolholders and spindle taper. Check for damaged toolholders before you put them back in the machine. A damaged toolholder can cause damage to the spindle taper and create tool runout or problems with finish.